

Person 4 — GeoAI / Urban Cadastral / GIS Processing

Intelligent Land Record Digitization, Validation & Urban Cadastral Mapping Platform |
Step-by-step implementation specification | Unified Master Blueprint v2.0

Assigned role: Person 4 — GeoAI / Urban Cadastral / GIS Processing

Document set: Unified Master Blueprint v2.0

Scope: SIH 12 + SIH 18 combined platform

Canonical rule: This document must not introduce a technology, endpoint, role, state, table, or handoff that conflicts with the Master Blueprint.

1. Mission

Own geospatial ingestion, CRS controls, tiling, segmentation/object detection, parcel/building/road extraction, polygonization, topology, comparison with existing cadastral layers and GIS exports. Treat all imagery-derived boundaries as preliminary until authorized verification.

2. Non-negotiable shared rules

- Use the canonical stack, API paths, entities, statuses, application roles and event names from Master Blueprint v2.0.
- Do not change another person's contract by silently renaming fields or endpoints.
- All work is project-scoped through `project_id`.
- All long-running work is asynchronous with job/workflow status.
- Original documents/source imagery remain immutable.
- Every AI output carries confidence, provenance, `model_version` and timestamp.
- Every critical mutation is auditable.
- Before handoff, run tests and provide evidence.

3. Exact handoff principle

A handoff is complete only when the receiving person can consume the documented output without guessing field names, endpoint names, status values, storage locations, or semantics.

4. Input contract

Asset	Required metadata/control
Drone imagery	CRS, bounds, GSD/pixel size, georeferencing, capture metadata
ORI / orthomosaic	CRS, transform, bounds, pixel size, NoData
DSM/DTM	CRS, elevation metadata, bounds, pixel size, NoData
Existing parcel GIS	CRS, schema, provenance, source authority/status
GT dataset	Location, labels, provenance, survey context
GNSS/CORS survey control	Coordinate reference/control metadata and provenance

5. Step-by-step GeoAI pipeline

- Register every asset under `project_id` and immutable storage.
- Validate raster/vector format, CRS/EPGS, extent, transform, GSD/pixel size, NoData, georeferencing and file integrity.
- Reject or quarantine invalid assets with explicit errors.
- Prepare spatially aware overlapping tiles while preserving affine transforms.
- Prepare labels for parcel boundaries, buildings, roads, pathways, water, vegetation, vacant land and other supported classes.
- Benchmark candidate segmentation/object-detection models on representative samples.
- Run inference on tiles.
- Merge tile outputs without losing world coordinates and remove edge artifacts.
- Polygonize masks into vector geometries.
- Apply controlled simplification, vertex snapping, sliver removal and geometry repair.
- Calculate geometry-derived area in an appropriate projected CRS.

- Run topology checks: validity, self-intersections, overlaps, unintended gaps and consistency.
- Compare predicted parcels with existing cadastral layers and produce change/link flags.
- Store confidence, source asset, model version, processing parameters and draft/approved status.
- Expose layers through Web-GIS and export GeoJSON/GeoPackage.
- Ensure draft edits do not overwrite original AI predictions.
- Emit geoai.completed with project_id, layer_ids, feature_counts and topology_error_count.

6. Canonical spatial entities

Entity	Core attributes
imagery_assets	id, project_id, uri, asset_type, crs, gsd, bounds_geom, capture_date, metadata_json, status
geoai_jobs	id, project_id, asset_id, job_type, status, parameters_json, metrics_json, model_version_id
parcels	id, project_id, source, survey_number, geom, area_m2, predicted_class, confidence, status, source_asset_id
buildings	id, project_id, geom, footprint_m2, confidence, status, source_asset_id
roads	id, project_id, geom, road_type, width_estimate, confidence, status, source_asset_id
land_use_features	id, project_id, class_name, geom, confidence, status, source_asset_id
topology_errors	id, project_id, feature_type, feature_id, error_type, geom, severity, status, details_json

7. Topology rules

- All geometries must be valid.
- Parcels must not self-intersect.
- Overlapping parcels are reported.
- Unintended gaps between adjacent parcels are reported.
- All project features use one declared CRS.
- Building polygons have valid rings and no extreme spikes.
- Road centerlines are connected where expected.
- Geometry area is positive and reproducible.

8. Model evaluation

Evaluation	Minimum evidence
Segmentation	Representative labeled validation set and documented metric(s)
Object detection	Representative precision/recall or task-appropriate metrics
Geometry	Polygon validity and topology outcomes
Area	Predicted vs manually measured/reference samples
Tile merging	No edge artifacts / coordinate drift
CRS	Multiple CRS/resolution test cases

9. GIS review rules

1. Open draft layer in Web-GIS.
2. Inspect imagery, predicted parcel, building, road and topology-error layers separately.
3. Inspect source asset metadata and CRS.

4. Review high-severity topology errors.
5. Edit only the draft version; preserve original prediction.
6. Save as a versioned correction with actor/reason.
7. Re-run geometry/topology validation.
8. Only publish when authorized workflow conditions are satisfied.

10. Handoff

- P3 receives geometry, area, identifiers and topology status for validation.
- P6 receives layers, feature attributes, topology errors and edit endpoints.
- P1 receives the canonical geoai.completed event.

11. Definition of done

- Multiple CRS/resolution inputs tested.
- Tile transforms preserved.
- Invalid geometries repaired or reported.
- Overlaps/gaps detected.
- GeoJSON/GeoPackage opens in QGIS.
- Draft edits do not overwrite AI predictions.
- Preliminary status is visible.
- GeoAI completion event passes.